Bhojan: Food & Grocery Platform

Project design for restaurants, tiffin services and small retail shops (kirana / general stores)
Working name · Design version 1.0 · 9 November 2026

Reference project reviewed: **Liquor Counter POS** (Fastify / Drizzle / React modular monolith). Assumptions are stated explicitly in Section 2. Tax rules, pricing and legal items must be confirmed with a Chartered Accountant before launch.

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1. Reuse Review of the Existing Project

The reference project is a liquor-retail POS and accounting platform: Fastify 5 + TypeScript API, Drizzle ORM on PostgreSQL (or embedded PGlite), zod validation, decimal.js money math, JWT auth with role and branch scoping, a React frontend, and Vitest/Playwright tests. It has an engine/service/routes split, domain events that drive double-entry accounting, a Tally export, bank reconciliation, dashboards, and KOT for bars.

> Basis: the project's own AGENTS.md (architecture, data model, domain rules). Individual source files were not all read, so the reuse estimates are judgment to confirm in a short spike.

Reuse map

Area	Verdict	Notes
Project structure (engine / service / routes)	Reuse as-is	Keeps business rules pure and unit-testable
Domain events and in-transaction handlers	Reuse as-is	Basis for stock, khata, accounting and notification side effects
Money math (decimal.js, numeric(12,2))	Reuse as-is	No floats for money
zod validation and error mapping	Reuse as-is	
JWT auth, role hierarchy, requireRole, per-request user re-read	Reuse as-is	Add PIN login for shared counter tablets
Idempotency via clientRef	Reuse as-is	Essential for offline sync
Bills never deleted; voided with reason	Reuse as-is	
Stock with stock_audit and guarded atomic SQL	Adapt	Add recipe / ingredient deduction for restaurants
Branch scoping and tenancy (syndicate to branch)	Adapt	Generalise to organisation to outlets
KOT (kitchen tickets)	Adapt	Becomes general kitchen tickets with a Kitchen Display
Tables and table sessions	Adapt	Restaurant module only

Area	Verdict	Notes
Accounting, Tally export	Adapt	Optional module, later phase
Bank reconciliation	Adapt	Optional, later phase
Dashboard engine (pure functions, server-side aggregation)	Reuse pattern	
Login throttling, seed data, dev-local PGlite runner	Reuse as-is	
Frontend components and design tokens	Adapt	Rebuild as offline-capable PWAs
Liquor rules (MRP tax-inclusive, excise quotas, pegs, sale types)	Remove or make configurable	
Counter sheets (per-type daily sheets)	Replace	Generic daily stock count
Supplier invoices and supplier schemes	Simplify	Purchase bills and GRN
Single language, server-first	Change	Add i18n and offline-first sync
Subscriptions, deliveries, routes, meal plans	New	
Customer notifications (WhatsApp / SMS)	New	

Estimate: about 40-50% of the backend foundation (auth, tenancy, engine pattern, events, money, stock audit, accounting) carries over.

2. Assumptions

#	Assumption	Why it matters
A1	Market is India first (INR, GST, UPI, FSSAI), with tax and currency kept configurable	Drives tax, payment and compliance design
A2	Owner-operators with 1-5 outlets are the main customer; chains come later	Keeps tenancy simple at first
A3	Staff are low-tech; must work on a low-cost Android phone and a cheap 58/80mm thermal printer	UI simplicity and offline requirement
A4	Internet is unreliable; power cuts happen	Offline-first is mandatory for billing
A5	WhatsApp is the main customer channel; SMS is the fallback	Notification design
A6	Tax rates are data, not code; a CA confirms rates per business type before go-live	Avoids hard-coded tax errors
A7	Small sellers may be unregistered or on the composition scheme and issue a Bill of Supply	Must be supported as a mode
A8	Team of 4-5 (2 backend, 2 frontend, 1 QA/design, part-time PM)	Roadmap and tech choices
A9	Payments are UPI, cash or card; card terminals integrated later	Limits MVP payment scope
A10	Personal data stays in India (DPDP Act readiness)	Hosting and backups

3. Target Users and Pain Points

User	Business	Key pain points	What they need most
Shop owner	Kirana	No daily profit view; stock runs out or expires; khata (credit) kept in a notebook	One screen: sales, profit, low stock, dues
Counter cashier	Kirana	Slow billing at rush; wrong prices; handwritten bills	Fast barcode/search billing, big buttons, offline

User	Business	Key pain points	What they need most
Restaurant owner	Restaurant	Order mix-ups; kitchen delays; wastage untracked	Table and KOT flow, kitchen screen, sales by item
Waiter	Restaurant	Walking back and forth; orders not reaching kitchen	Order on tablet, send to kitchen instantly
Kitchen staff	Restaurant / tiffin	Paper tickets lost; unclear quantities	Kitchen display with order queue
Tiffin owner	Tiffin	Counting meals daily; missed payments; pauses with no notice	Subscription manager, daily meal count, auto reminders
Tiffin customer	Tiffin	Forgets renewal; hard to skip a day	WhatsApp pause/skip, auto invoice, easy renewal
Delivery person	Tiffin / restaurant	Wrong address; no route; cash not reconciled	Ordered route, one-tap delivered, cash log
Accountant	All	Re-typing sales into Tally; GST filing errors	Clean exports, GST summary, purchase register

4. Functional Requirements

Priority: **M** = MVP, **L** = later phase.

4.1 POS / Billing

ID	Requirement	Pri
POS-1	Bill by barcode scan, product search (English and local-language name), or favourites grid	M
POS-2	Quantity, unit (kg/g/L/pcs) and variant (size, weight) selection	M
POS-3	GST computed per line from the product tax profile; invoice shows CGST/SGST/IGST	M
POS-4	Bill discount (amount or %) with manager approval above a limit	M
POS-5	Split payment (cash, UPI, card, khata)	M
POS-6	Print thermal receipt (58/80mm); share via WhatsApp/SMS link	M
POS-7	Hold and resume bills	M
POS-8	Void bill with reason; number kept, stock restored	M
POS-9	Return / credit note against an original bill	M
POS-10	Restaurant: item modifiers (extra spicy, no onion), course and seat	M (restaurant phase)
POS-11	Bill of Supply mode for unregistered sellers	M
POS-12	Price comes from server or local cache; never typed by cashier unless permitted	M

4.2 Inventory

ID	Requirement	Pri
INV-1	Product master: names (multi-language), SKU, barcode, HSN, tax slab, unit, MRP, cost	M
INV-2	Stock per outlet, auto-deducted on sale and restored on void	M
INV-3	Purchase entry / goods receipt (GRN) against supplier bill; updates cost	M
INV-4	Low-stock alerts and reorder suggestions	M
INV-5	Expiry dates and batch tracking (first-expiry-first-out) for perishables	M

ID	Requirement	Pri
INV-6	Stock adjustments with reason (damage, expired, sample, theft)	M
INV-7	Physical stock count with variance report	M
INV-8	Recipe-based ingredient deduction (restaurant)	L
INV-9	Raw-material stock for tiffin kitchen (daily purchase to consumption)	L
INV-10	Inter-outlet stock transfer	L
INV-11	Wastage log with cost impact	M

4.3 Orders

ID	Requirement	Pri
ORD-1	Order types: counter, takeaway, dine-in, delivery, subscription meal	M
ORD-2	Table management: floors, sections, occupancy, merge/split tables	L
ORD-3	Kitchen tickets routed to kitchen or bar stations; print or display	L
ORD-4	Kitchen Display System: new, cooking, ready, served	L
ORD-5	Order status visible to customer via WhatsApp link	L
ORD-6	Delivery order: address, phone, delivery charge, assign rider	L
ORD-7	Online ordering via shareable menu link (no app)	L
ORD-8	Aggregator intake (Swiggy, Zomato)	Out of scope initially

4.4 Customers and Khata

ID	Requirement	Pri
CUS-1	Customer profile: name, phone (unique per outlet), addresses, language	M
CUS-2	Khata (credit): limit, balance, statement, repayment	M
CUS-3	Purchase history and top customers	M
CUS-4	Automatic overdue reminder via WhatsApp	M
CUS-5	Loyalty points or cashback (configurable)	L
CUS-6	Consent record for marketing messages	M

4.5 Subscriptions (Tiffin)

ID	Requirement	Pri
SUB-1	Plans: meal type (lunch, dinner, both), days (all, weekdays, custom), period (weekly, monthly)	M
SUB-2	Menu calendar: owner sets daily or weekly menu; customers see it on WhatsApp	M
SUB-3	Customer pause, skip a day, or resume, before a cut-off time	M
SUB-4	Daily production sheet: meal count by type and by area, computed automatically	M
SUB-5	Packing labels with name, address and special instructions	M
SUB-6	Route planning: group by area, printable delivery sheet, optional map link	M
SUB-7	Delivery confirmation by rider (one tap); failure reason	M
SUB-8	Renewal reminders before expiry; auto-generated invoice; payment link	M
SUB-9	Proration for mid-cycle pause or cancellation (configurable)	L

ID	Requirement	Pri
SUB-10	One-off extra orders added to a subscription day	M

4.6 Staff

ID	Requirement	Pri
STF-1	Staff profiles, roles, active/inactive status, PIN login for shared tablets	M
STF-2	Shift open and close with opening and closing cash	M
STF-3	Simple attendance (clock in/out)	L
STF-4	Salary and advance tracking	L
STF-5	Activity log: who billed, voided, discounted what	M

4.7 Reporting

ID	Requirement	Pri
RPT-1	Owner dashboard: today vs yesterday, hourly sales, profit estimate, dues, low stock	M
RPT-2	Sales by day, item, category, cashier, payment mode	M
RPT-3	Stock report: value, movement, dead stock	M
RPT-4	GST summary (HSN-wise) and purchase register	M
RPT-5	Subscription report: active, paused, expiring, churn	M (tiffin)
RPT-6	Daily closing report (cash reconciliation)	M
RPT-7	Export to Excel/CSV (MVP); Tally (later)	M / L
RPT-8	Scheduled evening summary by WhatsApp or email	L

4.8 Payments

ID	Requirement	Pri
PAY-1	Cash with change calculation and drawer count	M
PAY-2	UPI dynamic QR per bill with amount; manual "payment received" confirmation	M
PAY-3	UPI webhook confirmation via payment gateway	M (tiffin) / L
PAY-4	Card via integrated terminal	L
PAY-5	Subscription payment via payment link (UPI autopay where supported)	L
PAY-6	Refunds recorded against bill; never silent	M
PAY-7	Supplier payments recorded (cash, bank transfer)	M

4.9 Notifications

ID	Requirement	Pri
NTF-1	Bill sent to customer via WhatsApp template or SMS	M
NTF-2	Khata reminder, subscription expiry reminder, daily menu broadcast	M
NTF-3	Order status updates to customer	L
NTF-4	Low-stock and daily summary alerts to owner	M
NTF-5	Honour opt-out and quiet hours; every send logged	M

ID	Requirement	Pri
NTF-6	Templates in customer's language	M

5. Non-Functional Requirements

Category	Requirement	Target
Performance	Bill completion (scan to print)	Under 2 s on 3G/4G; under 0.5 s offline
Performance	Product search	Under 150 ms for 50,000 products
Performance	Dashboard load	Under 2 s for a year of data
Availability	Billing works without internet	At least 8 hours (mandatory)
Availability	Cloud uptime	99.5% (99.9% in business hours)
Offline	Local cache of catalogue, prices, customers; offline bills queued and synced; stock can go negative locally and is flagged on sync	Mandatory
Offline	Idempotent sync via clientRef	Mandatory
Security	HTTPS only; passwords hashed (bcrypt or argon2); short-lived JWT with refresh token	Mandatory
Security	Role check on every endpoint; outlet scoping	Mandatory
Security	Audit log for voids, discounts, price changes, stock adjustments	Mandatory
Security	Encryption at rest for personal data (phone, address)	Mandatory
Security	Daily backups; monthly restore test	Mandatory
Privacy	DPDP Act: consent, purpose limitation, deletion on request, data in India	Mandatory
Scalability	One outlet: 1-3k bills/day. Design target: 500 outlets, 1M bills/month on one well-indexed Postgres	Design target
Scalability	Stateless API so it can run on several instances later	Design target
Multi-language	English, Hindi plus 2-3 regional languages at launch; i18n keys everywhere; product names stored per language	Mandatory
Multi-language	Receipts and WhatsApp templates localised	Mandatory
Localisation	Tax profiles as data (HSN, rate, cess, composition, inclusive or exclusive pricing)	Mandatory
Localisation	Indian rupee formatting (1,00,000); configurable currency later	Mandatory
Localisation	Fiscal year April to March, configurable	Mandatory
Localisation	FSSAI licence number on invoices (food businesses)	Mandatory
Localisation	e-invoicing (IRN) once turnover threshold is crossed	Later
Usability	New cashier bills unaided after 30 minutes of training; 48 px touch targets; portrait on a 6-inch phone	Mandatory
Accessibility	WCAG 2.1 AA contrast and focus; large-text option	Target
Observability	Structured logs, error tracking, uptime checks, sync-queue depth alert	Mandatory

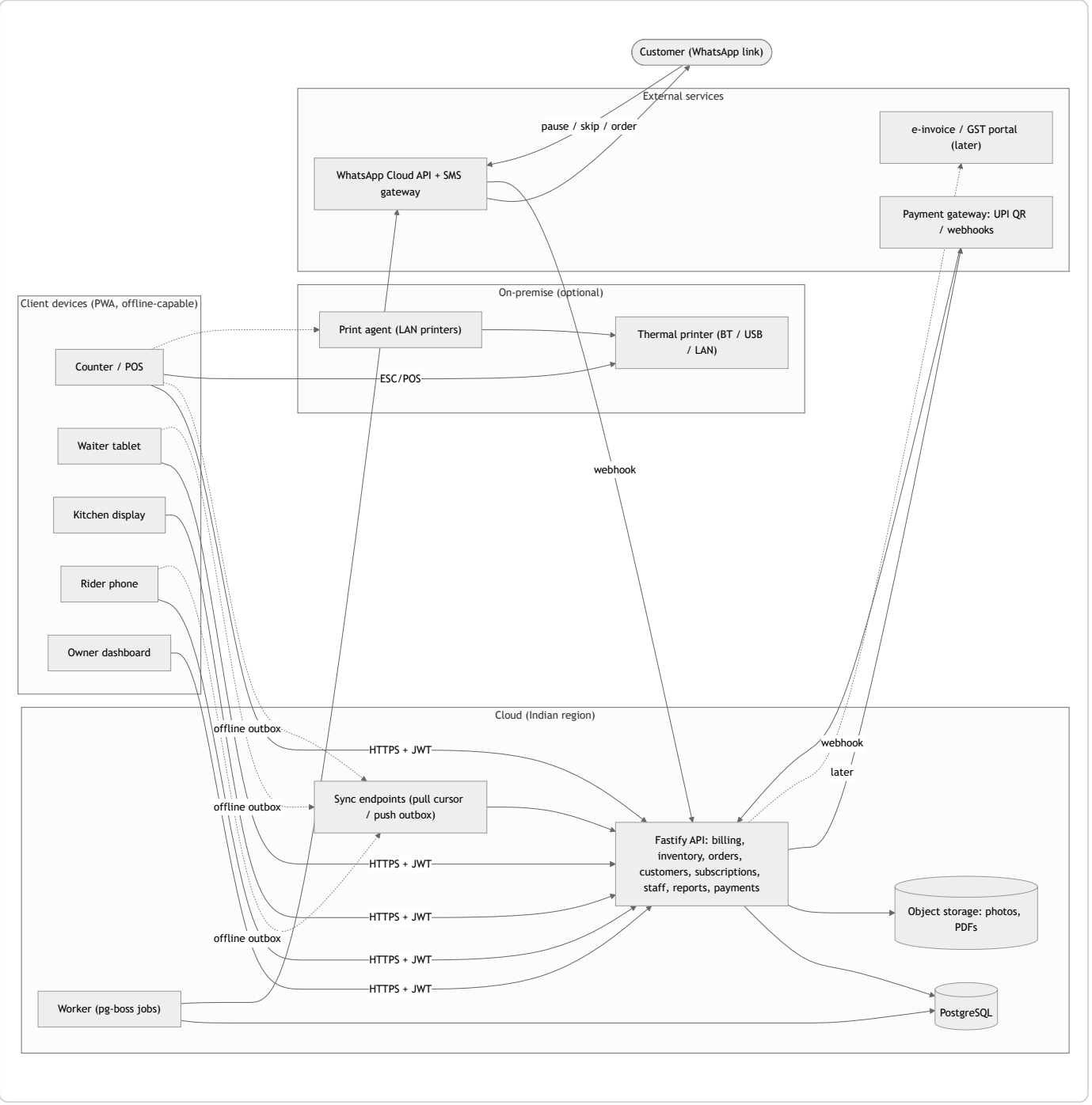
Category	Requirement	Target
Maintainability	Test coverage of engines (tax, discount, subscription scheduling, stock deduction) at 90% or more	Target

6. Recommended Tech Stack

Layer	Choice	Why
Language	TypeScript (strict) across backend and frontend	Shared types; one language for a small team
Backend	Node.js 22 + Fastify 5	Reused patterns; fast; low memory
Database	PostgreSQL 16 via Drizzle ORM	Strong transactions for money and stock; JSONB for flexible attributes
Local dev DB	PGlite (embedded)	Zero-install development
Validation / money	zod; decimal.js with numeric(12,2)	Reused
Auth	JWT (access + refresh) and PIN login for shared tablets	PIN fits shared-device reality
Background jobs	pg-boss (queue inside Postgres)	No Redis; retries for messages and reports
Frontend	React + Vite + Tailwind, built as a PWA	One codebase for cashier, waiter, rider, owner; installable; offline
Offline store	IndexedDB via Dexie.js + outbox pattern	Mature; simple sync queue
Sync	Pull changes by version cursor; push outbox with idempotent clientRef	Builds on existing idempotency design
i18n	i18next with ICU messages	Standard for Indian languages
Printing	ESC/POS over Bluetooth or USB via browser; optional local print agent	Cheap thermal printers; agent handles LAN printers
Notifications	WhatsApp Business Cloud API via provider; SMS gateway fallback	WhatsApp is the customer channel in India
Payments	UPI dynamic QR + gateway (e.g. Razorpay) for webhooks and payment links	Reliable confirmation without building payment integrations
File storage	S3-compatible object storage	Product photos, invoice PDFs
PDF	Server-side HTML to PDF (Playwright or light library)	One template engine
Hosting	Single VM (Docker Compose) in an Indian region, managed Postgres backups, CDN for PWA assets	Low cost; data residency; easy to operate
Observability	Pino logs, error tracking, uptime checker	Standard, low effort
Testing	Vitest (unit, integration on PGlite), Playwright (e2e including offline), k6 (load)	Reused
CI/CD	GitHub Actions: lint, test, build, container deploy	Standard
Mobile	PWA first; React Native only if hardware access requires it	Saves a large amount of effort

Why not microservices: a 4-5 person team needs one deployable modular monolith with clear module boundaries (the reference project's pattern). Split later only if a module such as notifications needs independent scaling.

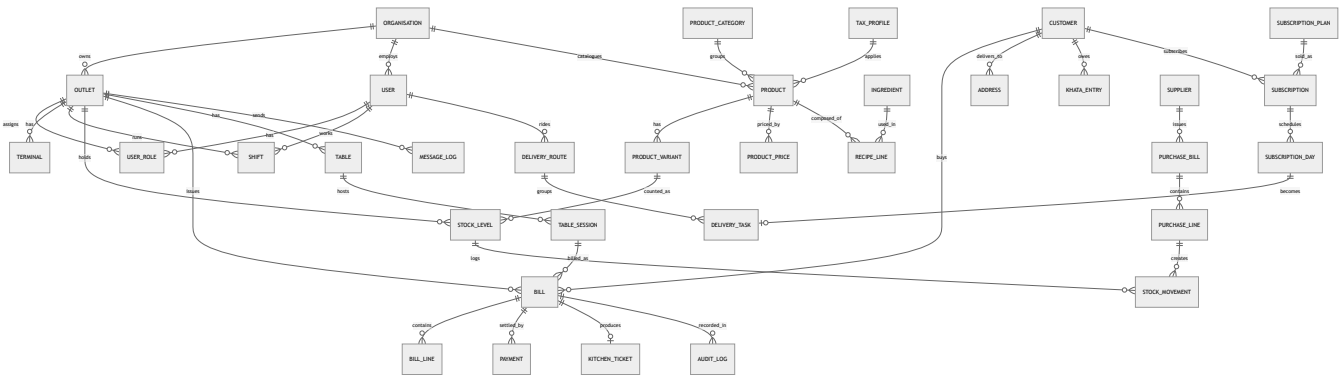
7. System Architecture



Key decisions: modular monolith with module folders (engine, service, routes); domain events between modules (for example *bill.paid* triggers stock deduction, khata update, accounting posting and a WhatsApp receipt), with money-critical handlers in the same transaction and notifications through the queue. Offline: each device keeps a local copy of catalogue and customers, writes bills to an outbox, and syncs with idempotent keys. Stock and credit limits are checked on the server at sync; conflicts are flagged for the manager rather than dropped. Every query is scoped by organisation and outlet.

8. Data Model

8.1 Main entities and relationships



8.2 Key tables

Table	Key fields	Notes
organisation	id, name, gstin, fssai_no, fiscal_year_start, business_type	One per owner account
outlet	id, org_id, name, address, gstin, invoice_prefix, business_type, currency, timezone	Business type sets defaults and enabled modules
tax_profile	id, hsn, cgst, sgst, igst, cess, inclusive_price, composition	Data, not code; per-outlet override
product	id, org_id, names (per language), sku, barcode, category_id, unit, tax_profile_id, type, attributes (JSON)	Attributes hold expiry flag, veg / non-veg, etc.
product_variant	id, product_id, label, mrp, cost, barcode	Size, weight, flavour
stock_level	outlet_id, variant_id, qty, reorder_level	Unique per outlet and variant
stock_movement	id, stock_level_id, qty_change, reason, ref_type, ref_id, batch, expiry, user_id, created_at	Append-only; every change has a reason
bill	id, outlet_id, number, type, status (OPEN / PAID / VOID), customer_id, subtotal, discount, tax, total, client_ref (unique), created_by, shift_id	Never deleted
bill_line	bill_id, variant_id, qty, unit_price, discount, tax_rate, taxable, tax, total, modifiers (JSON)	Price snapshot at sale time
payment	id, bill_id, mode (CASH / UPI / CARD / KHATA), amount, reference	Split payments
customer	id, org_id, name, phone (encrypted), language, consent_marketing, credit_limit, balance	Unique phone per organisation
khata_entry	id, customer_id, type, amount, balance_after, bill_id, client_ref	Append-only ledger
kitchen_ticket	id, outlet_id, bill_id, station, status, items (JSON), created_at	Source for the kitchen display
table / table_session	section, capacity, status / opened_at, closed_at, covers	Restaurant only
subscription_plan	id, name, meal_types, day_pattern, price, period, cutoff_time	e.g. Lunch, Mon-Sat, Rs 2,400 per month
subscription	id, customer_id, plan_id, start, end, status, address_id, prorate_rule	Active, paused, expired or cancelled

Table	Key fields	Notes
subscription_day	id, subscription_id, date, meal_type, status, menu_item_id, extra_of_id	Created ahead; changed by customer or owner
daily_menu	outlet_id, date, meal_type, items	Shown to customers and in production sheet
delivery_route / delivery_task	rider_id, date, status / sequence, status, cash_collected, failure_reason	Routes group tasks per rider
supplier / purchase_bill / purchase_line	GSTIN; number, date, total, tax; qty, cost, batch, expiry	Feeds stock and purchase register
staff_shift	user_id, terminal_id, opened_at, closed_at, opening_cash, closing_cash, expected_cash	Cash reconciliation
audit_log	id, actor_id, action, entity, entity_id, before, after, reason, at	Voids, discounts, price changes, adjustments
message_log	id, channel, template, to, status, provider_id, at	Only consent-checked sends
outbox_event	id, device_id, client_ref, payload, status, received_at	Idempotent sync receipt

9. Roles and Permissions

9.1 Roles

Role	Scope	Who it is
Owner	Organisation (all outlets)	Sees everything; sets prices and plans
Manager	One or more outlets	Runs daily operations; approves discounts, voids, adjustments
Cashier	One outlet, one shift	Bills, takes payment, opens and closes the drawer
Waiter	One outlet	Table orders; sends kitchen tickets
Kitchen	One station	Kitchen tickets only; no money screens
Delivery	Assigned routes	Own route; marks delivered and cash collected
Tiffin coordinator	One outlet	Subscriptions, meal counts, pauses, menu calendar
Accountant	Read-only, all outlets	Reports, GST summary, exports
Platform admin	Whole system (our team)	Support, tenant setup, software billing

9.2 Permission matrix

Y = allowed; L = allowed with approval or a limit; N = denied; - = not applicable.

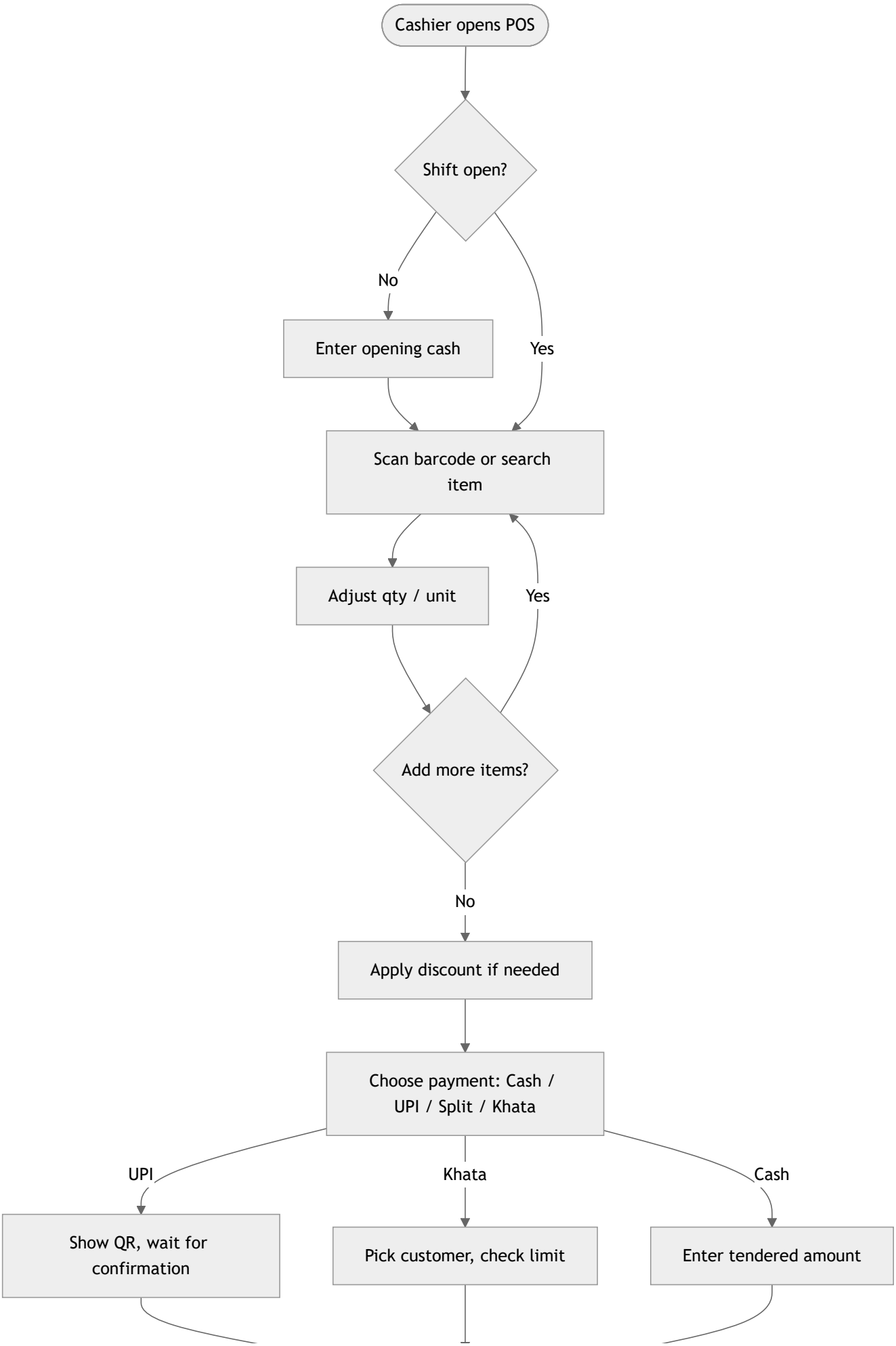
Action	Owner	Manager	Cashier	Waiter	Kitchen	Delivery	Tiffin coord.	Accountant
Create bill and take payment	Y	Y	Y	N	N	N	N	N
Send kitchen ticket	Y	Y	Y	Y	N	N	N	N
Update kitchen ticket status	Y	Y	N	N	Y	N	N	N
Apply discount	Y	Y	L (limit)	N	N	N	N	N
Void bill	Y	Y (reason)	L (request)	N	N	N	N	N

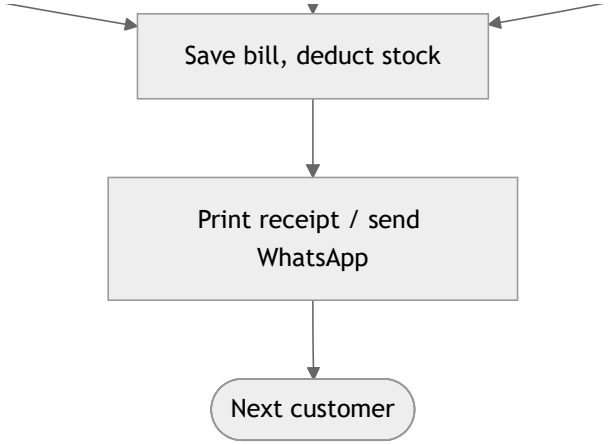
Action	Owner	Manager	Cashier	Waiter	Kitchen	Delivery	Tiffin coord.	Accountant
Change product price	Y	L	N	N	N	N	N	N
Add product or variant	Y	Y	N	N	N	N	N	N
Stock receipt (purchase)	Y	Y	L	N	N	N	N	N
Stock adjustment / wastage	Y	Y	L (wastage)	N	L (wastage)	N	L (raw material)	N
Set khata limits	Y	Y	N	N	N	N	N	N
Record khata repayment	Y	Y	Y	N	N	N	N	N
Manage subscriptions and menu calendar	Y	Y	N	N	N	N	Y	N
Plan routes, assign riders	Y	Y	N	N	N	N	Y	N
Mark delivered, cash collected	N	N	N	N	N	Y (own route)	Y	N
Open and close shift	Y	Y	Y (own)	N	N	N	N	N
Daily closing report	Y	Y	N	N	N	N	N	Y (view)
Profit and margin reports	Y	Y (no cost)	N	N	N	N	N	Y
Export GST / Tally	Y	Y	N	N	N	N	N	Y
Manage staff and roles	Y	L (below own role)	N	N	N	N	N	N
Audit log	Y	Y	N	N	N	N	N	Y

Rules everywhere: outlet scoping on every query; roles can grant only roles below their own; deactivation takes effect on the next request; cashier never sees cost price; every approval records who approved it.

10. Core User Flows

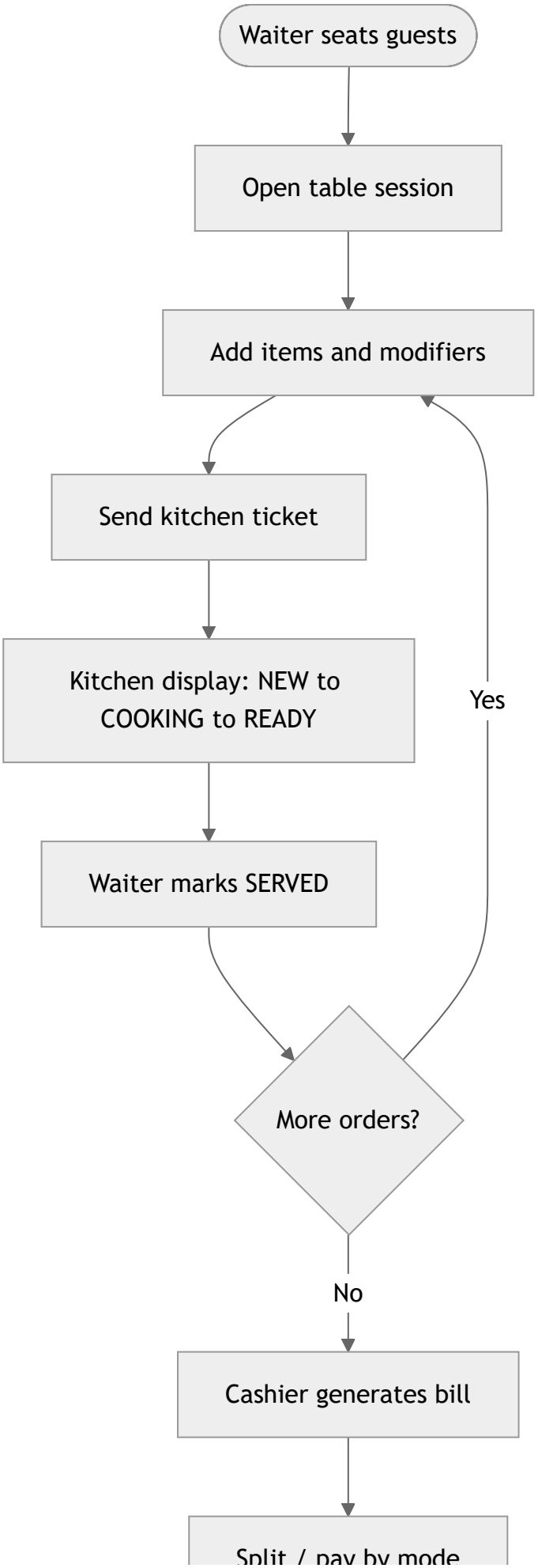
10.1 Kirana / general store

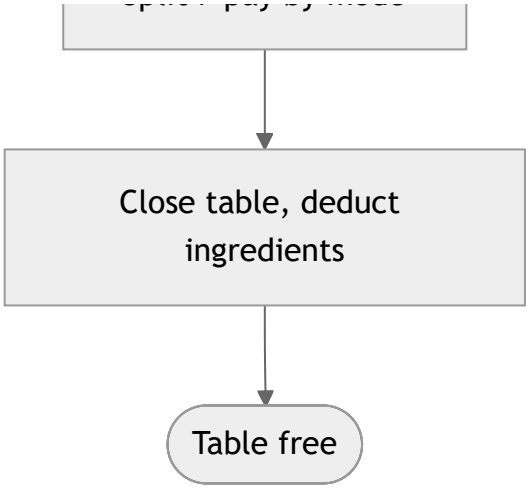

Cashier opens POS
Shift open?
No
Yes
Enter opening cash
Scan barcode or search item
Adjust qty / unit
Yes
Add more items?
No
Apply discount if needed
Choose payment: Cash / UPI / Split / Khata
UPI
Khata
Cash
Show QR, wait for confirmation
Pick customer, check limit
Enter tendered amount



- **Purchase:** supplier bill, then enter lines (scan, quantity, cost, expiry). Stock and cost are updated and the purchase register is written.
- **Day end:** cashier closes the shift with counted cash. The system shows expected vs actual; the manager reviews; the daily report goes to the owner.
- **Khata:** customer buys on credit, balance rises; repayment at the counter; automatic reminder after N days.

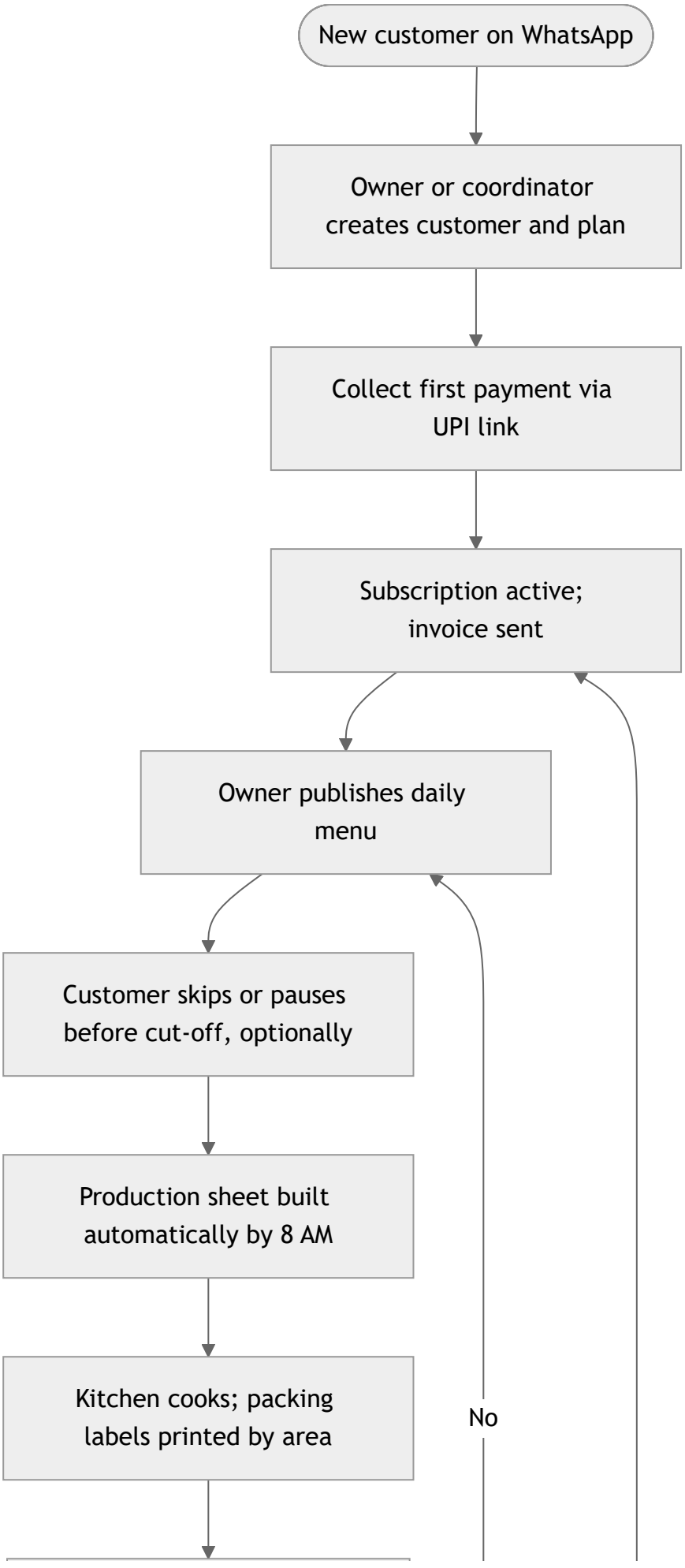
10.2 Restaurant

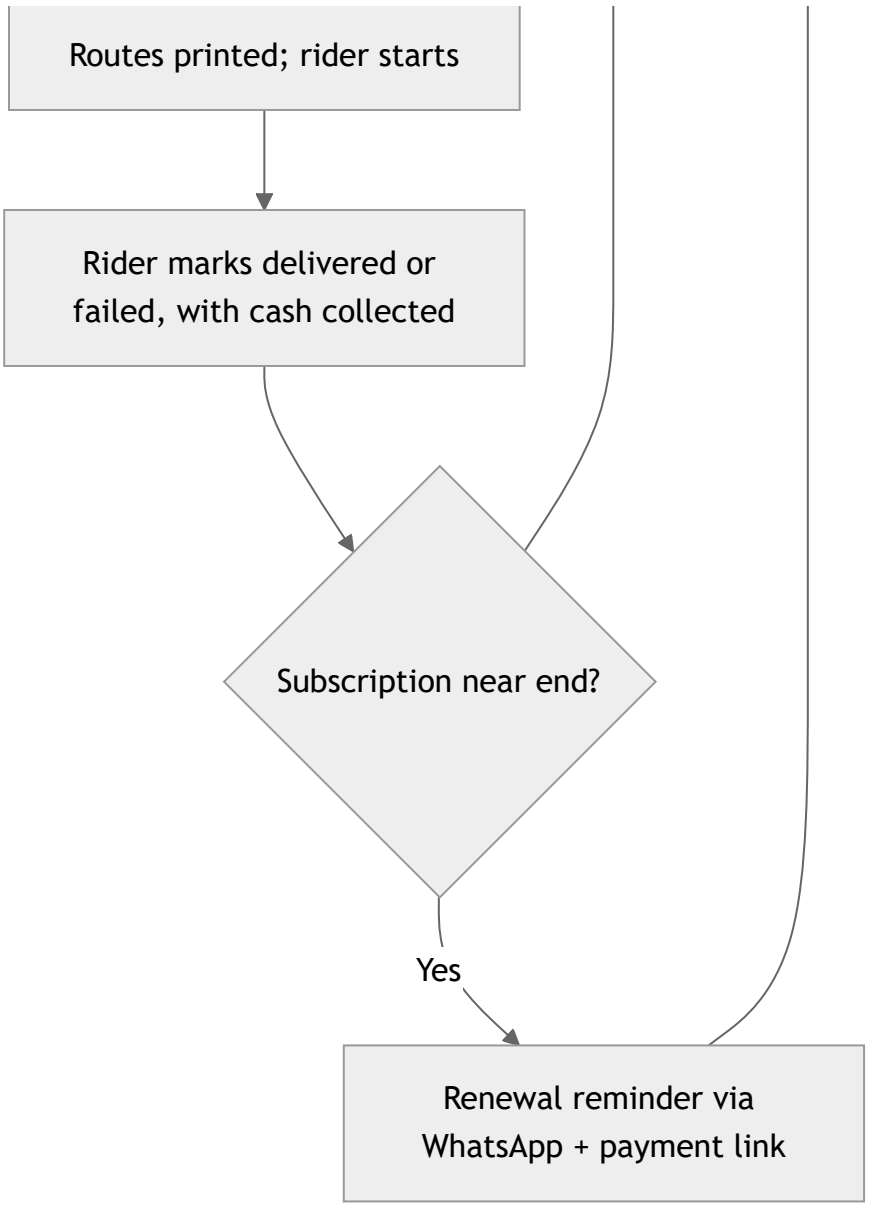

Waiter seats guests
Open table session
Add items and modifiers
Send kitchen ticket
Kitchen display: NEW to COOKING to READY
Waiter marks SERVED
More orders?
Yes
No
Cashier generates bill
Split / pay by mode



- **Walk-in / counter** uses the kirana flow, with a kitchen ticket printed for the kitchen.
- **Wastage:** kitchen logs wasted ingredients with a reason; it appears in the daily margin report.
- **Owner check:** dashboard shows top sellers, sales by hour, and cost of goods vs revenue.

10.3 Tiffin service


New customer on WhatsApp
Owner or coordinator creates customer and plan
Collect first payment via UPI link
Subscription active; invoice sent
Owner publishes daily menu
Customer skips or pauses before cut-off, optionally
Production sheet built automatically by 8 AM
Kitchen cooks; packing labels printed by area
No



- **Missed cut-off:** skip is charged or not, per the plan setting; the production sheet is already locked.
- **Day summary** at 9 PM to the owner: meals delivered, failed, skipped, cash collected, dues.

11. MVP vs Later Phases

Feature	MVP (Phases 1-2)	Later
Counter POS with GST invoice, barcode, split payment	Yes	
Offline billing and sync	Yes	
Stock with purchase entry, low-stock alerts, expiry	Yes	
Khata with reminders	Yes	
Shift closing and reports (sales, GST, stock)	Yes	
Thermal printing	Yes	
WhatsApp bill and reminders	Yes	
Multi-language UI (English, Hindi)	Yes	Regional languages in Phase 3

Feature	MVP (Phases 1-2)	Later
Tiffin subscriptions, menu calendar, meal count, routes	Phase 2	
Customer self-pause / skip via WhatsApp	Phase 2	
UPI payment-confirmation webhook	Phase 2	Auto-pay mandates
Restaurant tables, KOT, kitchen display		Phase 3
Modifiers, recipes, ingredient deduction		Phase 3
Card terminal integration		Later
Online menu link and ordering		Phase 4
Aggregator integration (Swiggy, Zomato)		Later, on demand
Multi-outlet and stock transfer	Basic in Phase 4	Full version later
Tally export		Phase 4
e-invoice (IRN)		When required
Loyalty points		Phase 4
Staff attendance and payroll		Later
Native mobile apps		Only if needed

12. Development Roadmap

Team of 4-5 with sustained delivery. Durations are estimates to validate in Phase 0.

Phase	Duration	Scope	Exit criteria
0. Discovery and design	3 weeks	Interview 5-8 shop owners across the three types; confirm tax rules with a CA; finalise data model; clickable prototype of POS and tiffin daily sheet	Design signed off; pilot shops identified (2 kirana, 2 tiffin, 1 restaurant)
1. Core platform and kirana POS	10-12 weeks	PIN auth, organisation and outlet, catalogue, tax profiles, POS billing, split payment, printing, offline cache and sync, stock and purchases, khata, shift close, dashboard and GST reports, WhatsApp bill	Pilot kirana shop bills for 2 weeks with no internet-related loss; daily closing matches cash on 95%+ of days
2. Tiffin subscriptions	8 weeks	Plans, customers, menu calendar, pause/skip, production sheet, labels, routes, rider PWA, renewals and payment links, day summary	Pilot tiffin runs a full month on the system; no missed deliveries from the list; renewals collected via link
3. Restaurant	8-10 weeks	Tables, KOT, kitchen display, modifiers, recipes and ingredient deduction, wastage, regional-language UI	Pilot restaurant runs a weekend service on the kitchen display; order-to-kitchen under 5 seconds
4. Scale and integrations	8-10 weeks	Multi-outlet, stock transfer, Tally export, UPI webhook for all, loyalty, online menu link, e-invoice readiness	5 paying outlets; monthly subscription billing working
5. Hardening and launch	Ongoing	Load test, security review, DPDP compliance review, backup restore drill, onboarding videos, support process	Public launch with support playbook

Sequencing: kirana first because it reuses the most of the existing project and proves the core (billing, stock, offline, GST). Tiffin second for the most recurring revenue with less complexity than tables and kitchen tickets. Restaurant last because it has the most moving parts.

13. Key Risks and Mitigations

#	Risk	Likelihood	Impact	Mitigation
R1	Offline sync creates duplicate bills or wrong stock	Medium	High	Idempotent clientRef; server assigns bill numbers on sync with per-device number ranges; stock conflicts flagged, never silently overwritten
R2	Tax rules change or are wrong for a business type	High	High	Tax as data, reviewed by a CA per business type; versioned tax profiles with effective dates; test cases per slab
R3	Shop owners do not adopt (low tech, paper habit)	High	High	Pilot 5 shops in Phase 1; 30-minute onboarding; print a receipt that looks like the old bill; WhatsApp support; few mandatory fields
R4	Cheap phones and printers behave inconsistently	High	Medium	Device test matrix (5 phones, 4 printers) in Phase 0; prefer Bluetooth printers with known ESC/POS profiles; print agent fallback
R5	WhatsApp template approval delays or account restriction	Medium	Medium	Approved template categories only; SMS fallback; honour opt-out; no bulk spam
R6	Customer data privacy complaints (DPDP)	Medium	High	Consent capture; encrypt phone and address; deletion flow; data in India; privacy notice during onboarding
R7	Scope creep into restaurant and tiffin at once	High	High	Phase gates with exit criteria; one business type per phase; feature flags per business type
R8	Subscription billing errors (proration, missed skips)	Medium	High	Subscription engine as pure, heavily tested functions; daily reconciliation report; dry run before cut-off
R9	UPI gap: customer paid, shop unsure	Medium	Medium	Dynamic QR with amount and bill reference; manual confirmation with reason; webhook in Phase 2; daily UPI reconciliation
R10	Poor stock data (wrong counts, no barcodes)	High	Medium	Bulk import from spreadsheet; quick "add product on the fly" form; cycle counts instead of one annual count
R11	Single point of failure (one VM, one database)	Low-Medium	High	Daily encrypted off-site backups; monthly restore drill; standby replica once revenue justifies it
R12	Team too small for three business types plus offline plus i18n	High	High	Shared core engine; PWA rather than native apps; i18n keys from day one; contract a designer for Phase 0
R13	Competition from free or cheap POS apps	Medium	Medium	Differentiate on tiffin subscriptions and khata, which generic POS apps handle poorly; price for small shops

14. Decisions Needed Before Phase 0 Closes

1. **Name and brand** (Bhojan is a working name).
2. **First launch market** (single state or pan-India); affects regional languages and tax examples.
3. **Pricing model** (monthly per outlet, or per-feature modules).
4. **Hosting preference** (cloud provider and region; whether on-premise is ever required).
5. **Payment gateway** (e.g. Razorpay) and **WhatsApp provider** (direct Meta API or a business solution provider).
6. **Pilot shops:** at least 5 across the three business types, willing to test weekly.
7. **Team:** confirm headcount and whether a designer is available.

Summary

The reference project gives a solid, reusable backend foundation: modular structure, role-based access, tenancy, money math, stock audit, idempotency and accounting. The new product keeps that core, replaces liquor-specific rules with configurable tax and product data, and adds three things the reference lacks: **offline-first operation**, **multi-language support**, and **subscription and delivery workflows** for tiffin. Build kirana billing first, then tiffin, then restaurant, with each phase gated on pilot-shop results.

Bhojan platform design v1.0 · Prepared 9 November 2026 · Tax, legal and pricing items are assumptions pending professional review.